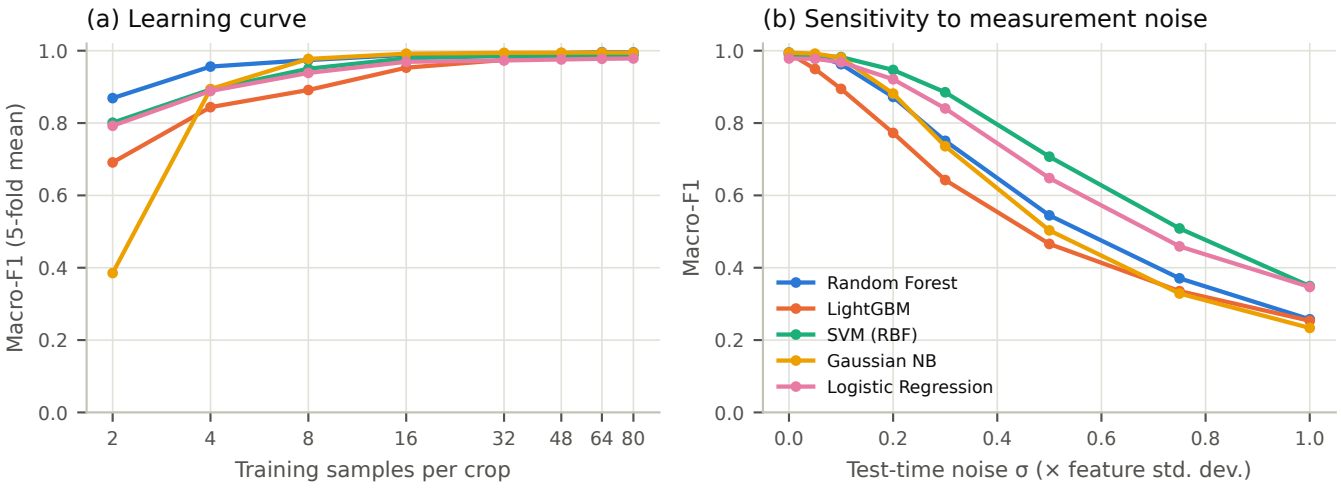
(a) Learning curve
Macro-F1 (5-fold mean)
1.0
0.8
0.6
0.4
0.2
0.0
2
4
8
16
32
48
64
80
Training samples per crop
(b) Sensitivity to measurement noise
Macro-F1
Random Forest
LightGBM
SVM (RBF)
Gaussian NB
Logistic Regression
0.0
0.2
0.4
0.6
0.8
1.0
Test-time noise σ (× feature std. dev.)